



Test report

Generated By: Sentina AI

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Report Type: sustainability

This report was automatically generated by the Sustainability Management System.

SUMMARY

Overview of this report based on the configuration selected in the Sustainability dashboard.

Custom Notes / Remarks

testin

Report Configuration

- **Date Range:** 2026-03-29 to 2026-04-05
- **Zone(s):** zoneA
- **Facility / Hall:** HZA03
- **Aggregation Level:** Hourly

Included Sections

- Occupancy & Events Overview
- Energy Consumption
- Environmental Metrics
- Definitions and Calculation Logic

Occupancy & Events Overview

No occupancy data available for the selected filters.

Try changing date range / zones / facilities.

Message

No occupancy data available for the selected filters.

Energy Consumption

No energy data available for the selected filters.

Try changing date range / zones / facilities.

Message

No HVAC energy (hvacEnergyKWh) data available for the selected filters.

Environmental Metrics

No environmental data available for the selected filters.

Try changing date range / zones / facilities.

Message

No rows available after filtering.

Definitions and Calculation Logic

Reference guide for energy, occupancy, environmental, and sustainability calculations.

Metric Definitions

Definitions and interpretation of sustainability, energy, occupancy, and environmental metrics.

Metric	Definition	Interpretation
Capacity	Configured maximum occupancy capacity of the hall.	Used as the baseline for utilization and peak-capacity comparisons.
Average Occupancy	Mean number of occupants across the selected period.	Higher values indicate more sustained hall usage.
Peak Occupancy	Highest observed occupancy in any time bucket.	Represents the strongest crowd load observed.
Peak % Capacity	Peak occupancy expressed as a percentage of configured hall capacity.	Values close to 100% indicate the hall reached full designed occupancy.
Event Count	Number of events associated with the hall or selected grouping in the period.	Higher values indicate broader event activity within the selected scope.
Event IDs	Identifiers of events linked to the selected hall, peak window, or grouped record.	Used to trace which specific events contributed to the observed result.
Total Energy Consumption (kWh)	Total energy consumed within the selected time period.	Higher values indicate greater energy usage.
HVAC Energy Consumption (kWh)	Energy consumed specifically by HVAC systems.	Indicates cooling and heating load contribution to total energy use.
Carbon Emissions (kg CO2)	Estimated carbon emissions associated with energy consumption.	Higher values indicate greater environmental impact.
Peak Energy Usage	Maximum energy consumption observed within a time bucket.	Represents peak load conditions.
Average Energy Usage	Average energy consumption across the selected period.	Indicates the general consumption trend over time.
kWh / Occupant	Energy consumption normalized by occupancy level.	Helps compare energy usage more fairly across different crowd levels.
Occupancy-Adjusted Energy	Energy consumption normalized by occupancy levels.	Used to compare halls or intervals with different people counts.
Energy Intensity	Energy consumed per unit occupancy or per square meter.	Lower values indicate better efficiency.
Energy Efficiency Score	Normalized score reflecting how efficiently energy is used relative to demand.	Ranges from 0 to 1. Higher values indicate better efficiency.
Normalized Energy Consumption	Average energy consumption scaled relative to the maximum observed value.	Ranges from 0 to 1 and supports score-based comparison.
Normalized Carbon Emissions	Average carbon emissions scaled relative to the maximum observed value.	Ranges from 0 to 1 and supports sustainability scoring.

Metric	Definition	Interpretation
Sustainability Score	Composite sustainability score combining energy efficiency and carbon impact.	Higher values indicate more sustainable operation.
Maximum Efficiency Condition	Theoretical condition where efficiency is highest and sustainability score approaches 1.	Represents the most favorable energy-performance condition.
Average Outdoor Temp (°C)	Mean outdoor temperature across the selected grouping or time window.	Provides environmental context for cooling and energy demand.
Average Indoor Temp (°C)	Mean indoor temperature across the selected grouping or time window.	Used to assess internal thermal conditions and comfort performance.
Temp Δ (°C)	Difference between indoor and outdoor temperature.	Shows the thermal gap being maintained by the venue environment.
Humidity (%)	Relative humidity level within the selected hall or time bucket.	Used as a direct environmental condition indicator affecting comfort.
Average Humidity (%)	Mean relative humidity across the selected scope.	Used to compare humidity conditions across halls or periods.
Comfort Index	Composite comfort score derived from indoor temperature, humidity, and related environmental conditions.	Higher values indicate better environmental comfort.
Comfort Status	Categorical interpretation of the comfort index.	Labels conditions such as acceptable or uncomfortable.
Worst Comfort	Lowest comfort index observed for a hall during the selected period.	Represents the most uncomfortable observed condition.
Worst Time	Time bucket in which the lowest comfort index was observed.	Identifies when the weakest environmental condition occurred.
Top Discomfort Moment	A ranked time window with the lowest comfort or highest thermal stress.	Highlights the most critical discomfort condition observed.

Calculation Logic

How sustainability metrics are derived.

Metric	Formula / Logic
Peak % Capacity	$(\text{Peak Occupancy} / \text{Capacity}) \times 100$
Total Energy Consumption (kWh)	$\text{sum}(\text{energy_kwh})$
HVAC Energy Consumption (kWh)	$\text{sum}(\text{hvac_energy_kwh})$
Carbon Emissions (kg CO2)	$\text{energy_kwh} \times \text{emission_factor}$
kWh / Occupant	$\text{energy_kwh} / \text{max}(\text{current_occupancy}, 1)$
Occupancy-Adjusted Energy	$\text{energy_kwh} / \text{max}(\text{current_occupancy}, 1)$
Energy Intensity	$\text{energy_kwh} / \text{hall_area}$ OR $\text{energy_kwh} / \text{occupancy}$
Energy Efficiency Score	$1 - \text{normalized_energy_consumption}$
Normalized Energy Consumption	$\text{avg_energy} / \text{max}(\text{avg_energy across dataset})$
Normalized Carbon Emissions	$\text{avg_carbon} / \text{max}(\text{avg_carbon across dataset})$
Sustainability Score	$0.5 \times (1 - \text{normalized_energy}) + 0.5 \times (1 - \text{normalized_carbon})$

Metric	Formula / Logic
Maximum Efficiency Condition	Occurs when energy consumption and emissions are at minimum
Average Outdoor Temp (°C)	average(outdoor_temp_c)
Average Indoor Temp (°C)	average(indoor_temp_c)
Temp Δ (°C)	Indoor Temperature - Outdoor Temperature
Average Humidity (%)	average(humidity_pct)
Comfort Index	Derived comfort model
Comfort Status	Rule-based classification from Comfort Index thresholds
Worst Comfort	min(comfort_index)
Worst Time	argmin(comfort_index)
Top Discomfort Moment	Rank by lowest comfort_index

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